

Dual Spatial Formulation and Reproducing Kernel Hilbert Space Theory for the Krafft Sieve

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Abstract

This working note details a reformulation of the variational optimization of spectral sieve quotients. By projecting the generalized eigenvalue problem from the $2^{w(n)}$ -dimensional character coefficient space down to the spatial evaluation interval $\mathcal{A}_n$ of size $12n + 4$, we establish a dual formulation. Under this duality, the optimization is shown to be equivalent to the minimization of a diagonal penalty matrix restricted to the range of an evaluation operator. We derive a closed-form product kernel for the full power-set basis, as well as a Newton-identity-based dynamic programming recurrence for restricted correlation depths. Finally, we frame this dual system within the theory of Reproducing Kernel Hilbert Spaces (RKHS), laying out potential avenues for analytical operator-theoretic bounds.

1 Introduction and the Primal Formulation

The primal formulation of the Krafft sieve optimization (outlined in Section 5 of the manuscript) seeks a weight function $W(x) \geq 0$ supported on the interval $\mathcal{A}_n = [6n^2 - 2n, 6n^2 + 10n + 3]$ to minimize the sieve quotient:

$$\mu(n, W) = \frac{\sum_{x \in \mathcal{A}_n} c(x)W(x)}{\sum_{x \in \mathcal{A}_n} W(x)} \quad (1)$$

where $c(x)$ is the global additive penalty function counting the modular alignment hits. To ensure non-negativity without imposing infinitely many linear constraints, we restrict $W(x)$ to be the square of a polynomial $P(x)$ over a basis of cosine products:

$$P(x) = \sum_{S \subseteq \{1, \dots, w(n)\}} \lambda_S B_S(x), \quad B_S(x) = \prod_{i \in S} \cos\left(\frac{6\pi x}{p_i}\right) \quad (2)$$

Here, $w(n) = |\mathcal{P}_n|$ is the number of primes in the sieve window, and the basis dimension is $D = 2^{w(n)}$. Squaring $P(x)$ yields the weight function $W(x) = P(x)^2$ on $\mathcal{A}_n$. The total mass S_1 and the penalized mass S_2 are then quadratic forms:

$$S_1(n, W) = \lambda^T B \lambda, \quad S_2(n, W) = \lambda^T A \lambda \quad (3)$$

where A and B are $D \times D$ matrices defined by:

$$B_{S,T} = \sum_{x \in \mathcal{A}_n} B_S(x) B_T(x), \quad A_{S,T} = \sum_{x \in \mathcal{A}_n} c(x) B_S(x) B_T(x) \quad (4)$$

Minimizing the sieve quotient corresponds to finding the minimum generalized eigenvalue of the system:

$$A\lambda = \mu B\lambda \quad (5)$$

Solving (5) directly is computationally impossible for large n because the matrix dimension $D = 2^{w(n)}$ grows exponentially. For example, when $n = 199$, $w(n) = 194$, and the matrix dimension is $2^{194} \approx 2.4 \times 10^{58}$.

2 The Dual Spatial Reformulation

Let $N = |\mathcal{A}_n| = 12n + 4$ be the cardinality of the evaluation interval. Let M be the $N \times D$ evaluation matrix whose entries are:

$$M_{x,S} = B_S(x) \quad \text{for } x \in \mathcal{A}_n, S \subseteq \{1, \dots, w(n)\} \quad (6)$$

We can express the mass matrix B and the penalty matrix A in terms of M as:

$$B = M^T M, \quad A = M^T C M \quad (7)$$

where C is the $N \times N$ diagonal matrix with diagonal entries $C_{x,x} = c(x)$.

The rank of M is at most $\min(N, D)$. Since $N \ll D$ for large n , the matrices A and B are extremely rank-deficient, with a null space of dimension at least $D - N$. Any vector in the kernel of M (so $M\lambda = 0$) lies in the kernel of both A and B , yielding trivial 0/0 quotients. The non-trivial spectrum resides in the orthogonal complement of the kernel of M , which is the range of M^T .

Let $P = M\lambda \in \mathbb{R}^N$ be the vector of polynomial values evaluated over the interval $\mathcal{A}_n$. The Rayleigh quotient transitions directly to the spatial domain:

$$R(\lambda) = \frac{\lambda^T M^T C M \lambda}{\lambda^T M^T M \lambda} = \frac{P^T C P}{P^T P} \quad (8)$$

Let $V = \text{range}(M) \subseteq \mathbb{R}^N$ be the subspace spanned by the evaluated basis functions. Minimizing $R(\lambda)$ over $\lambda \in \mathbb{R}^D \setminus \ker(B)$ is equivalent to minimizing the spatial quotient over the subspace V :

$$\mu_{\min}(n) = \min_{P \in V \setminus \{0\}} \frac{P^T C P}{P^T P} \quad (9)$$

This is a standard Rayleigh quotient minimization of a diagonal matrix C restricted to a subspace V .

To solve (9), we construct an orthonormal basis for V . Consider the $N \times N$ symmetric, positive semidefinite kernel matrix:

$$K = M M^T \quad (10)$$

Let $K = U \Sigma U^T$ be the spectral decomposition of K . Let $r = \text{rank}(K) \leq N$ be the number of positive eigenvalues of K . Let U_r be the $N \times r$ matrix whose columns are the eigenvectors corresponding to these positive eigenvalues. Because $\text{range}(K) = \text{range}(M) = V$, the columns of U_r form an orthonormal basis for V .

Theorem 2.1 (Dual Sieve Theorem). *The non-trivial generalized eigenvalues of the primal system $A\lambda = \mu B\lambda$ are exactly the eigenvalues of the $r \times r$ projected matrix:*

$$\tilde{C} = U_r^T C U_r \quad (11)$$

In particular, the minimum sieve quotient is the smallest eigenvalue of $\tilde{C}$:

$$\mu_{\min}(n) = \lambda_{\min}(\tilde{C}) \quad (12)$$

Proof. Let $M^T C M \lambda = \mu M^T M \lambda$ with $\mu \neq 0$ and $M \lambda \neq 0$. Let $P = M \lambda$. Multiplying the equation by M on the left gives:

$$M M^T C P = \mu M M^T P \implies K C P = \mu K P \quad (13)$$

Since $P \in \text{range}(M) = \text{range}(K)$, we can write $P = U_r \gamma$ for a unique non-zero vector $\gamma \in \mathbb{R}^r$. Substituting this yields:

$$K C U_r \gamma = \mu K U_r \gamma \quad (14)$$

Expressing $K = U_r \Sigma_r U_r^T$, where Σ_r is the $r \times r$ diagonal matrix of positive eigenvalues:

$$U_r \Sigma_r U_r^T C U_r \gamma = \mu U_r \Sigma_r U_r^T U_r \gamma \quad (15)$$

Since $U_r^T U_r = I_r$, this simplifies to:

$$U_r \Sigma_r (U_r^T C U_r) \gamma = \mu U_r \Sigma_r \gamma \quad (16)$$

Multiplying by U_r^T on the left and using $U_r^T U_r = I_r$:

$$\Sigma_r (U_r^T C U_r) \gamma = \mu \Sigma_r \gamma \quad (17)$$

Since Σ_r is invertible (containing only positive eigenvalues), multiplying by Σ_r^{-1} gives:

$$(U_r^T C U_r) \gamma = \mu \gamma \quad (18)$$

This is a standard eigenvalue problem of dimension $r \leq N$. The steps are fully reversible for any non-zero spatial vector, establishing the result. $\square$

3 The Closed-Form Kernel and Symmetric Polynomials

The dual formulation reduces the problem to diagonalizing the $N \times N$ matrix $K = M M^T$. The entries of K are:

$$K_{x,y} = \sum_S B_S(x) B_S(y) = \sum_S \prod_{i \in S} \cos\left(\frac{6\pi x}{p_i}\right) \cos\left(\frac{6\pi y}{p_i}\right) \quad (19)$$

The complexity of this construction depends on the choice of the basis family.

3.1 The Unrestricted Power-Set Basis

If the polynomial basis contains all subsets $S \subseteq \{1, \dots, w(n)\}$, the sum over the power set factors into a product:

$$K_{x,y} = \prod_{p \in \mathcal{P}_n} \left(1 + \cos\left(\frac{6\pi x}{p}\right) \cos\left(\frac{6\pi y}{p}\right) \right) \quad (20)$$

Equation (20) is a closed-form expression. It can be computed for all $x, y \in \mathcal{A}_n$ in $O(N^2 \cdot w(n))$ operations. Diagonalizing K takes $O(N^3)$ operations. This makes solving the exact, unrestricted $2^{w(n)}$ -dimensional optimization problem computationally cheap.

3.2 Restricted Correlation Depths

If the basis is restricted to subsets S of size at most k , the kernel becomes:

$$K_{x,y}^{(k)} = \sum_{j=0}^k e_j(\mathbf{a}) \quad (21)$$

where $\mathbf{a} \in \mathbb{R}^{w(n)}$ has components $a_i = \cos(6\pi x/p_i) \cos(6\pi y/p_i)$, and $e_j(\mathbf{a})$ is the j -th elementary symmetric polynomial.

We can compute the sum of these symmetric polynomials efficiently using a dynamic programming recurrence. Let $dp[i][j]$ be the j -th elementary symmetric polynomial of the first i components of $\mathbf{a}$. The recurrence relation is:

$$dp[i][j] = dp[i-1][j] + a_i dp[i-1][j-1] \quad (22)$$

with boundary conditions $dp[i][0] = 1$ and $dp[i][j] = 0$ for $j > i$.

To compute this on a computer, we can maintain a single vector of size $k+1$ and update it in-place for each prime:

```

Initialize  $dp = [1, 0, \dots, 0]$  of length  $k+1$ 
for each prime  $p_i \in \mathcal{P}_n$  do
     $a_i = \cos(6\pi x/p_i) \cos(6\pi y/p_i)$ 
    for  $j = k$  down to 1 do
         $dp[j] \leftarrow dp[j] + a_i \cdot dp[j-1]$ 
    end for
end for
return  $\sum_{j=0}^k dp[j]$ 

```

This algorithm computes the restricted kernel entry $K_{x,y}^{(k)}$ in $O(w(n) \cdot k)$ operations, yielding an overall kernel assembly time of $O(N^2 \cdot w(n) \cdot k)$.

4 Connection to Reproducing Kernel Hilbert Spaces

The kernel matrix $K_{x,y}$ is the discrete evaluation of a symmetric, positive semidefinite kernel function. This kernel defines a Reproducing Kernel Hilbert Space (RKHS), denoted by $\mathcal{H}_K$, of functions on the interval $\mathcal{A}_n$.

Every function $P \in \mathcal{H}_K$ can be written as $P(x) = \sum_S \lambda_S B_S(x)$ for some coefficients λ . The RKHS norm of P corresponds to the L^2 norm of its coefficients:

$$\|P\|_{\mathcal{H}_K}^2 = \sum_S \lambda_S^2 \quad (23)$$

The evaluation matrix M acts as an evaluation operator mapping the coefficient space ℓ^2 to the spatial space $L^2(\mathcal{A}_n)$. The adjoint operator M^* maps spatial functions back to the coefficient space. The kernel matrix $K = MM^T$ is the spatial representation of the operator MM^* .

Within this Hilbert space structure:

1. The subspace $V = \text{range}(M)$ is the closed subspace of $L^2(\mathcal{A}_n)$ containing all functions that can be represented as "smooth" character combinations.
2. The sieve quotient is the Rayleigh quotient of the multiplication operator C (where $CP(x) = c(x)P(x)$) restricted to the subspace $V \subseteq L^2(\mathcal{A}_n)$.

This RKHS perspective opens up several theoretical paths:

- **Analytical Bounds:** By modeling the continuous limit of the kernel function as $n \rightarrow \infty$, we can apply Mercer’s theorem to decompose $K(x, y)$ into continuous eigenfunctions. This could allow us to analytically bound the spectrum of the projected multiplication operator without relying on discrete numerical computation.
- **Optimal Sieve Weights:** The optimal spatial weight function is $P_{\text{opt}}(x)^2$, where P_{opt} is the eigenfunction corresponding to the smallest eigenvalue of the projected operator $\tilde{C}$. The shape of this eigenfunction represents the optimal compromise between localized support on $\mathcal{A}_n$ and destructive interference at the hit frequencies.

4.1 Lean 4 Formalization of Mercer’s Theorem

To ground these theoretical pathways in machine-checked code, the Reproducing Kernel Hilbert Space structure can be formalized using Mathlib’s active RKHS library. Leveraging discussions with Mathlib contributors, Mercer’s theorem for a vector-valued RKHS can be stated using standard Mathlib idioms (specifically, the continuous L^2 operator $Lp\ V\ 2\ \mu \rightarrow L[\mathbb{K}]Lp\ V\ 2\ \mu$ and the complete orthonormal basis class `HilbertBasis`).

Below is the formal Lean 4 theorem statement defining the integral operator and the corresponding spectral decomposition:

```
import Mathlib.Analysis.InnerProductSpace.Reproducing
import Mathlib.Analysis.InnerProductSpace.HilbertBasis
import Mathlib.MeasureTheory.Function.LpSpace

open MeasureTheory Matrix HilbertBasis

variable {K : Type*} [RCLike K]
variable {X : Type*} [TopologicalSpace X] [CompactSpace X] [MeasurableSpace X] [BorelSpace X]
variable {V : Type*} [NormedAddCommGroup V] [InnerProductSpace K V] [CompleteSpace V]
variable {H : Type*} [NormedAddCommGroup H] [InnerProductSpace K H] [CompleteSpace H]
variable [RKHS K H X V]
variable (μ : Measure X) [IsFiniteMeasure μ]

-- The continuity assumption wrapped as a Fact
variable [hK : Fact (Continuous (fun (p : X × X) ↦ kernel H p.1 p.2))]

-- The integral operator as a continuous linear map from L^2 to L^2
def integralOperator : Lp V 2 μ → L[K] Lp V 2 μ := sorry

/-- Mercer’s Theorem (Refined with Mathlib idioms): -/
theorem mercer_theorem :
  ∃ (ι : Type*) [Countable ι]
    (b : HilbertBasis ι K (Lp V 2 μ))
    (σ : ι → ℝ),
    (∀ i, 0 ≤ σ i) ∧
    -- Eigenvalue equation: T_K(b_i) = σ_i b_i
    (∀ i, integralOperator μ (b i) = (σ i : K) · (b i)) ∧
    -- The kernel decomposition holds in the RKHS (using evaluation)
    (∀ (x y : X) (v : V),
      kernel H x y v = ∑' i, (σ i : K) · ⟨v, (b i : X → V) y⟩_K · (b i : X → V) x) := by
  sorry
```